

Generation Interconnection System Impact Study Report

For Queue Project Q22-1983

Waterford Power Llc 500 kV

December 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1983. The Interconnection Customer proposes a 435.1 MW natural gas combined cycle facility interconnecting at the Waterford Power Llc 500 kV substation in the Ohio area. The request was received on January 13, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 699 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$72,749,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Brentwood / Muskingum Riv 230 KV
Contingency: Tidd / Muskingum Riv 500 KV line
Loading: 106.0% (Rating: 580 MVA)
DFAX: 9.8%
Pre-project loading: 98.6%
2. Facility: Mine Road / Brentwood 230 KV
Contingency: Tidd / Muskingum Riv 500 KV L
Loading: 105.4% (Rating: 573 MVA)
DFAX: 9.8%
Pre-project loading: 98.0%
3. Facility: Oreg 3 Inc / Brooks Hill 230 KV
Contingency: Muskingum Riv / Tidd 500 KV Line
Loading: 105.3% (Rating: 405 MVA)
DFAX: 9.5%
Pre-project loading: 95.2%
4. Facility: Wes Bellaire / Mine Road 230 KV
Contingency: Tidd / Muskingum River 500 KV
Loading: 105.2% (Rating: 566 MVA)
DFAX: 9.8%
Pre-project loading: 97.6%
5. Facility: Waterford Power Llc / Muskingum River 230 KV line
Contingency: Myrtle-Date Substation / Waterford Power Llc 230 KV Line
Loading: 104.9% (Rating: 460 MVA)
DFAX: 7.0%
Pre-project loading: 98.3%
6. Facility: Brooks Hill / Myrtle-Date 230 KV
Contingency: Tidd / Muskingum Iver 500 KV
Loading: 102.9% (Rating: 437 MVA)

DFAX: 9.5%
Pre-project loading: 93.5%

7. Facility: Oreg 3 Inc 345 kV/230 kV Auto
Contingency: Muskingum River / Tidd 500 KV line
Loading: 101.2% (Rating: 310 MVA)
DFAX: 6.8%
Pre-project loading: 91.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brentwood / Muskingum River Substation 230 KV	Reconductor line	\$19,431,000	100% (\$19,431,000)
Mine Road / Brentwood 230 KV	Reconductor line	\$10,651,000	100% (\$10,651,000)
Oreg 3 Inc / Brooks Hill 230 KV	Reconductor line	\$2,519,000	85% (\$2,144,000)
Wes Bellaire / Mine Road 230 KV	Reconductor line	\$10,267,000	100% (\$10,267,000)
Waterford Power Llc / Muskingum Riv 230 KV Line	Reconductor line	\$3,188,000	100% (\$3,188,000)
Brooks Hill / Myrtle-Date 230 KV	Reconductor line	\$13,504,000	85% (\$11,491,000)
Oreg 3 Inc 345 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$18,380,000	85% (\$15,577,000)
	Total allocated to project		\$72,749,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.